

Enough about probability (for now). Let's turn now to some results from contemporary physics which will be important to the argument which follows.

A first question: what does the theory provide by physics include?

“The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.” (Hawthorne & Isaacs, “Fine-tuning fine-tuning”)



These 'dimensional constants' will be our focus.

One such constant is the cosmological constant, which measures the energy density of empty space.

The cosmological constant is just a number which (as far as we know) could have different values consistent with the laws of physics.

One thing that the standard model of physics gives us is a measure of how likely it is, given the laws of the nature, that the fundamental constants (like the cosmological constant) would fall in a certain range.

We can make certain plausible assumptions about what it would take for life to have evolved. For example, if there were nothing but hydrogen, it is hard to see how life could have evolved. If there were no planets, it is hard to see how life could have evolved.

Given assumptions such as these, we can look at what the standard model of physics tells us about how likely it is that, for example, the cosmological constant has a value which would permit the evolution of life.

“Physicists have determined the (approximate) values of the fundamental constants by measurement.

(There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

This is the claim that that the cosmological constant having a life-supporting value, given the laws of nature, is very low.

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

The claim is that, according to current physics, the probability of the cosmological constant falling in a life-supporting range, given the laws of nature, is $\frac{1}{10^{120}}$.

This is the claim that that the cosmological constant having a life-supporting value, given the laws of nature, is very low.

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

Let's now ask how to turn these facts about current physics and probability theory into an argument for the existence of God.

Much as we considered two different hypotheses about the number of balls in the urn, so we can consider two different hypotheses about the origins of the universe.

Let the **design hypothesis** be the hypothesis that the universe was created by an intelligent designer. Let the **non-design hypotheses** be the hypothesis that it was not.

Let our **evidence** (e) be the fact that the cosmological constant has a value in the life-permitting range.

Then what do we know about relevant probabilities?

$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$

$$Pr(e \mid \text{design}) = 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h)*P(e|h)}{P(e)}$$

With this information in hand, figuring out the probability of the non-design hypothesis given our evidence is a matter of just plugging in the numbers.

$$Pr(\text{non-design}) = 0.5$$

$$Pr(\text{design}) = 0.5$$

$$\Pr(e) = 0.25$$

$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

It is difficult to think about numbers as large as the denominator of this fraction. But to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now consider the odds of winning Powerball one trillion times in a row. Call that a “super Powerball.”

Now consider the odds of winning a super Powerball one trillion times in a row. Call that a “super duper Powerball.”

Now consider the odds of winning a super duper Powerball one trillion times in a row.

The odds of this happening are about $1 / 10^{44}$ — so much, much higher than the odds of the universe being life-permitting by chance.

This means that if you simply take the physics at face value, and begin by assigning a probability of 0.5 to the non-design hypothesis, you should think that the chances of the non-design hypothesis being true are vastly lower than the chances of winning a super duper powerball a trillion times in a row.

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1 minus the very small number we were just discussing:

$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This about as close to certainty as it is possible to get.

Instead, I am going to explore the possibility that one might respond to this argument by questioning some of the assumptions about probabilities that the argument makes.



fine-
tuning of the
universe

A solid red circle with a subtle drop shadow, centered on a white background. Inside the circle, the text "Bayes' theorem" is written in a white, sans-serif font, centered both horizontally and vertically.

Bayes'
theorem



the
argument



objections

What we want to know is how changes in those assumptions about probability affect the probability that one should assign to the design hypothesis.

$$A = P(\text{design}) =$$

$$B = P(\text{life} \mid \text{design}) =$$

$$C = P(\text{life} \mid \text{non-design}) =$$

$$P(\text{design} \mid \text{life}) =$$





0.5

0

.

5

1



10120

Let's go back to the view that the prior probability of design is 0.5, rather than 1 in a million. Might one challenge the probability assigned to life given non-design?

Sure, that probability was based on the findings of contemporary physics. But we are surely not **certain** that physics is correct about this. What if there is, say, a 10% chance that physics was fundamentally wrong about some major things, and that the odds of the constants having life supporting values in that case is not $\frac{1}{10^{120}}$, but rather something like 0.5?

$$0.5 * 0.5$$

$$0.5 * 0.5 + (1 - 0.5) * \frac{1}{10^{120}}$$

Enough about probability (for now). Let's turn now to some results from

follows.

contemporary physics which will be important to the argument which

One such constant is the cosmological constant, which measures the

energy density of empty space.

could have different values consistent with the laws of physics.

The cosmological constant is just a number which (as far as we know)

likely it is, given the laws of the nature, that the fundamental constants

One thing that the standard model of physics gives us is a measure of how

(like the cosmological constant) would fall in a certain range.

life to have evolved. For example, if there were nothing but hydrogen, it is

to see how life could have evolved.

We can make certain plausible assumptions about what it would take for

hard to see how life could have evolved. If there were no planets, it is hard

constant has a value which would permit the evolution of life.

of physics tells us about how likely it is that, for example, the cosmological

Given assumptions such as these, we can look at what the standard model

supporting value, given the laws of nature, is very low.

This is the claim that the cosmological constant having a life-



omplimentary constant falling in a life-supporting range, given the laws of

The claim is that, according to current physics, the probability of the

nature, is

1

10120

This is the claim that the cosmologist has a life-

supporting value, given the law of nature, is very low.

Let's now ask how to turn these facts about current physics and

probability theory into an argument for the existence of God.

of the universe.

Much as we considered two different hypotheses about the number of balls

in the urn, so we can consider two different hypotheses about the origin

hypothesis that it was not.

created by an intelligent designer. Let the non-design hypothesis be the

Let the design hypothesis be the hypothesis that the universe

value in the life-permitting range.

Let e be the fact that the cosmological constant has a



$$\Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$



$$Pr(e | \text{design}) \equiv 0.5$$

numbers.

design hypothesis given evidence is that just plugging in the

With this information in hand, figuring out the probability of the non-



$$Pr(\text{non-design}) = 0.5$$



$$Pr(\text{design}) = 0.5$$



$$Pr(e) = 0.25$$



$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now



consider the odds of winning Powerball one trillion times in a row. Call that a “super

It is difficult to think about numbers as large as the denominator of this fraction. But

PowerBar!

Now consider the odds of winning a super Powerball one trillion times in a row. Call

that a 'super duper Powerball.'

odd of the universe being life-permitting by chance.

The odds of this happening are about $1/10^{44}$ — so much, much higher than the

Now consider the odds of winning a super Powerball one trillion times in a row.

superduperpowerball a trillion times in a row.

This means that if you simply take the physics at face value, and begin by assigning a

the non-design hypothesis being true are vastly lower than the chances of winning a

probability of 0.5 to the non-design hypothesis, you should think that the chances of

minus the very small number we're just discussing:

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1.



$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

to this argument by questioning some of the assumptions about

Instead, I'm going to explore the possibility that one might respond

probabilities that the argument makes.

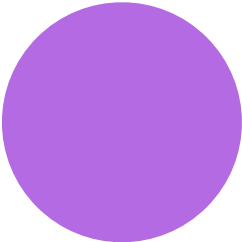


A solid red circle serves as the background for the text.

Bayes'
theorem



the
argument



What we want to know is how changes in these assumptions about

probability affect the probability that one should assign to the design

hypothesis.

$$B \equiv P(\text{life} \mid \text{design}) \equiv$$

$A \equiv P(\text{design}) \equiv$

$$C \equiv P(\text{life} \mid \text{non-design}) \equiv$$

1



10120

Let's go back to the view that the prior probability of design is 0.5, rather

than 1 in a million. Mightn't he probably as good as have been given

non-design?

, but rather something like 0.5?

validates in that case is not

Sure, that probability was based on the findings of contemporary physics.

some major things, and that the cost of having life support

But we are surely not certain that physics is correct about this. What if

there is, say, a 10% chance that physics was fundamentally wrong about

1



10120

0

.

5



0

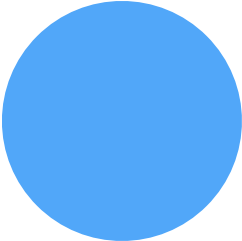
.

5

$$0.5 * 0.5 + (1 - 0.5) *$$

1

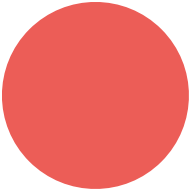
10120





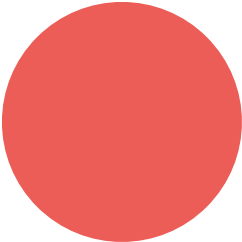


fine-
tuning of the
universe



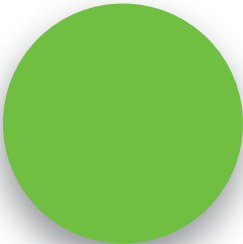


fine-
tuning of the
universe









nature, is

cosmological constant falling in a life-supporting range, giving the laws of

The claim is that, according to current physics, the probability of the



objections

Sure, that probability was based on the findings of contemporary physics.

, but rather something like 0.5?

some major things, and that the odds of the constants having life supporting

there is, say, a 10% chance that physics was fundamentally wrong about

But we are sure not certain that physics is correct about this. What if

values in that case is not